

BEAM ANALYSIS by the STIFFNESS METHOD

Only Pin Supports and Point Loads are included here. Distributed Loads are not included yet.
Does not include provisions for applied moments yet.

Matrix programs / routines based on ORIGIN =1.

version: beta v.2026.07.20

PROBLEM DESCRIPTION

Problem description & details summary can be input in this box.

Problem Description (short): *problem* := “single span, 3 point loads”

Begin beam positions:
(always ZERO) $x_0 := 0 \text{ ft}$

End beam position (beam length): $x_{end} := 50 \text{ ft}$

Total Beam Length $L_{beam} := x_{end} = 50 \text{ ft}$

Number (qty) of result points
printed along elements: $n_r := 10$

INPUT - MATERIAL

Material Type: *material* := steel ▾

Modulus of Elasticity: $E := material_2 = (2.9 \cdot 10^4) \text{ ksi}$

Section Description: *section* := “W14x61”

Moment of Inertia of Beam: $I := 640 \text{ in}^4$

INPUT - SUPPORT LOCATION

All supports assumed to be pins at this time.

Support Position Table

x_{supp} (ft)
0
25
50

Support Positions:

$$x_{supp}^T = [0 \ 25 \ 50] \text{ ft}$$

Number of Supports:

$$n_{supp} := \text{rows}(x_{supp}) = 3$$

INPUT - POINT LOAD TABLE

Enter any number of point loads. Negative forces acting downward.

Working Values

Truck Axle Load(s) - UNITLESS:

$$P_a := 8$$

$$x_1 := 7.5$$

$$x_2 := x_1 + 6$$

Point Load Input Table:

(can insert rows below to add more loads)

x_{Pload} (ft)	P_x (lbf)	M_x (lbf · ft)
40	-5000	0

check_point := "check to make sure all loads are less than the beam length"

INPUT - DISTRIBUTED LOAD TABLE

Enter any number of distributed loads, maintaining the (R rows x columns)

DISTRIBUTED LOADS ARE NOT INCLUDED IN THE ANALYSIS YET!

Distributed Load Input Table:

(can insert rows below to add more loads)

x_{wstart} (ft)	x_{wend} (ft)	w_{start} $\left(\frac{\text{kip}}{\text{ft}}\right)$	w_{end} $\left(\frac{\text{kip}}{\text{ft}}\right)$
0	4	-1	-1

NOT INCLUDED IN ANALYSIS YET!!!

 $check_dist :=$ “check to make**WORKING EQUATIONS - NON-PRINTING AREA****REACTION OUTPUT EQUATIONS - NON-PRINTING AREA****SHEAR OUTPUT EQUATIONS - NON-PRINTING AREA****MOMENT OUTPUT EQUATIONS - NON-PRINTING AREA****OUTPUT - SUPPORT REACTIONS** $Reactions_{node} := [\text{“Node”} \text{ “position”} \text{ “vert. reaction”}]$

Node, position, and vert. reaction:

$$Reactions_{node}^T = \begin{bmatrix} 1 & 2 & 4 \\ 0 \text{ } m & 7.62 \text{ } m & 15.24 \text{ } m \\ -1.868 \cdot 10^3 \text{ } N & (1.263 \cdot 10^4) \text{ } N & (1.148 \cdot 10^4) \text{ } N \end{bmatrix}$$

$$ReacOut_{node}^T = [1 \ 2 \ 4]$$

$$ReacOut_{pos}^T = [0 \ 25 \ 50] \text{ } ft$$

$$ReacOut_{force}^T = [-0.42 \ 2.84 \ 2.58] \text{ } kip$$

OUTPUT - SHEAR FORCE

THIS ANALYSIS DOES NOT YET INCLUDE ANY DISTRIBUTED LOADS.

Shear forces at nodes:

$$V_{node} := \begin{bmatrix} \text{"Node"} & \text{"position"} & \text{"Left force"} & \text{"Right force"} \end{bmatrix}$$

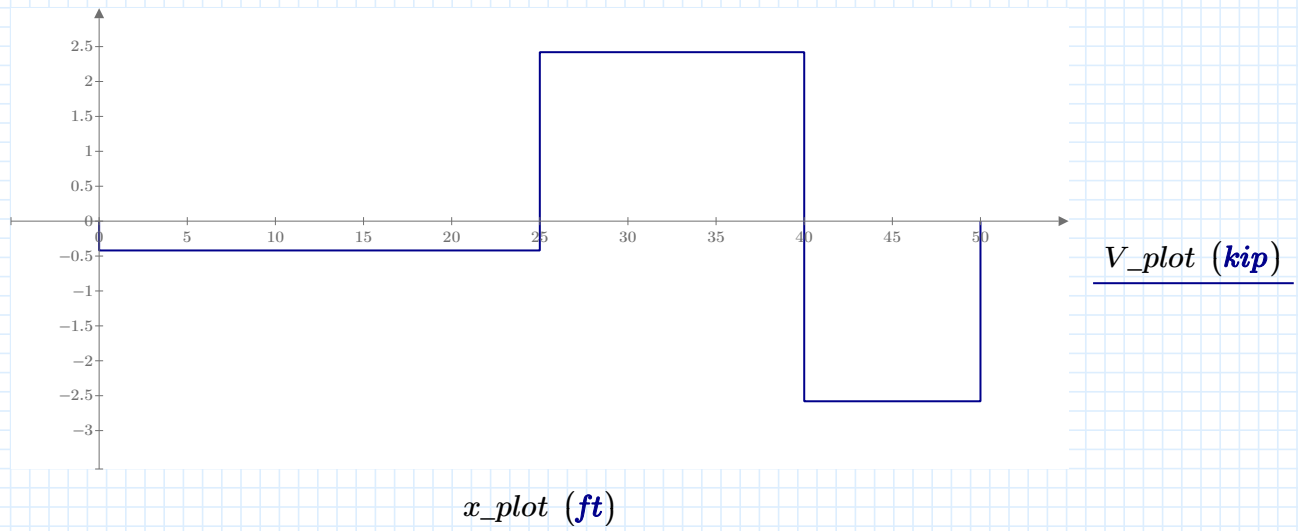
$$V_{node} = \begin{bmatrix} 1 & 0 \text{ m} & 0 \text{ N} & -1.868 \cdot 10^3 \text{ N} \\ 2 & 7.62 \text{ m} & -1.868 \cdot 10^3 \text{ N} & (1.076 \cdot 10^4) \text{ N} \\ 3 & 12.192 \text{ m} & (1.076 \cdot 10^4) \text{ N} & -1.148 \cdot 10^4 \text{ N} \\ 4 & 15.24 \text{ m} & -1.148 \cdot 10^4 \text{ N} & 0 \text{ N} \end{bmatrix}$$

$$Vout_{node}^T = [1 \ 2 \ 3 \ 4]$$

$$Vout_{pos}^T = [0 \ 25 \ 40 \ 50] \text{ ft}$$

$$Vout_{left}^T = [0 \ -0.42 \ 2.42 \ -2.58] \text{ kip}$$

$$Vout_{right}^T = [-0.42 \ 2.42 \ -2.58 \ 0] \text{ kip}$$



OUTPUT - MOMENT

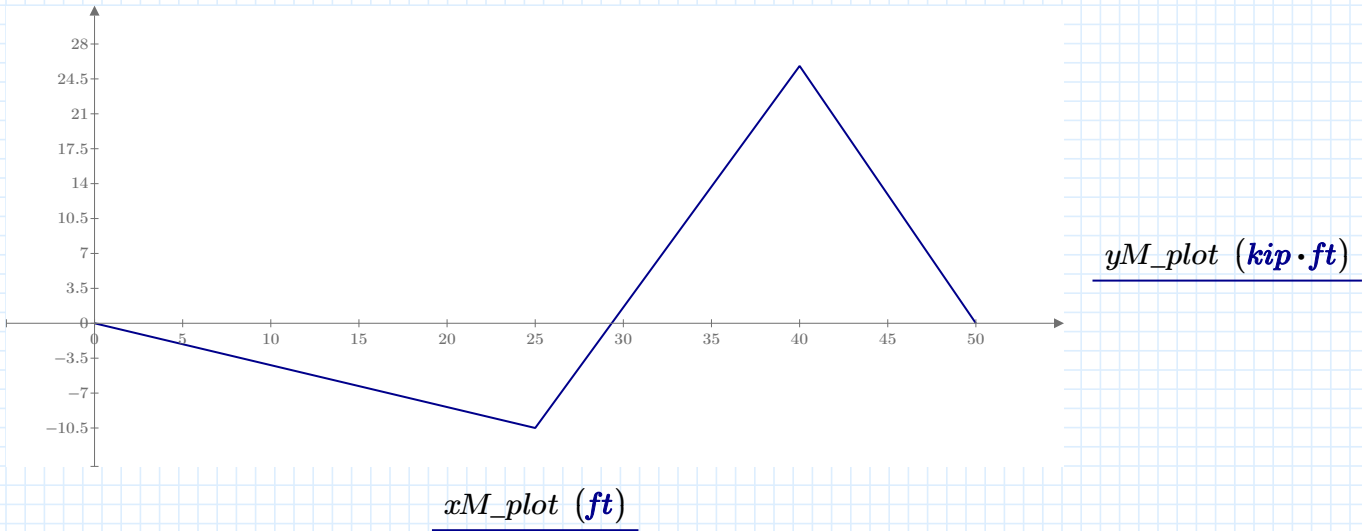
THIS ANALYSIS DOES NOT YET INCLUDE ANY DISTRIBUTED LOADS.

Specific / sample point: $x_s := 6.25 \text{ ft}$

Moment value at specified point: $M_{pnt}(x_s) = -2.625 \text{ kip} \cdot \text{ft}$

Maximum Moment: $M_{max} := \max(M_{out}) = 25.8 \text{ kip} \cdot \text{ft}$

Minimum Moment: $M_{min} := \min(M_{out}) = -10.5 \text{ kip} \cdot \text{ft}$



INPUT SUMMARY

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

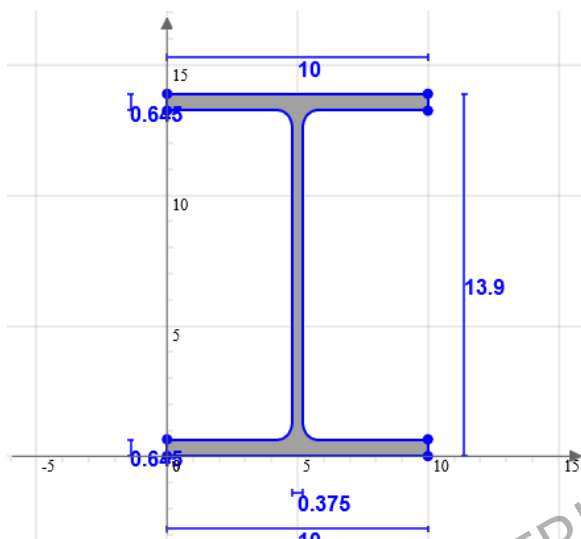
Supports

Support Type	Location
Pinned	0 ft
Roller	50 ft
Roller	25 ft

Loads

Load Type	Location	Magnitude	Load Case
Point Load	40 ft	-5000 lb	DL

Beam Section



Geometric Properties		
A	17.9	in ²
C _z	5	in
C _y	6.95	in

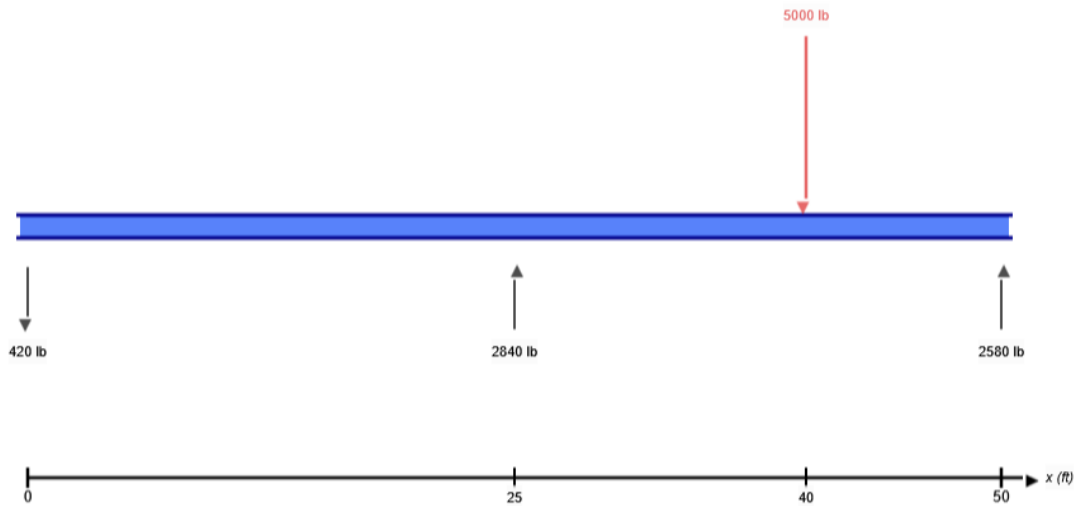
Bending Properties		
I _z	640	in ⁴
I _y	107	in ⁴

Shear Properties		
A _z	11.277	in ²
A _y	4.99	in ²

Torsion Properties		
J	2.19	in ⁴
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft ³)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.044	L/5648
Custom Stress Limit	PASS	39 ksi	0.086	3.362 ksi
Material Yield	PASS	36 ksi	0.093	3.362 ksi
Material Strength	PASS	58 ksi	0.058	3.362 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
0	0	-420	0
25	0	2840	0
50	0	2580	0

Force Extremes

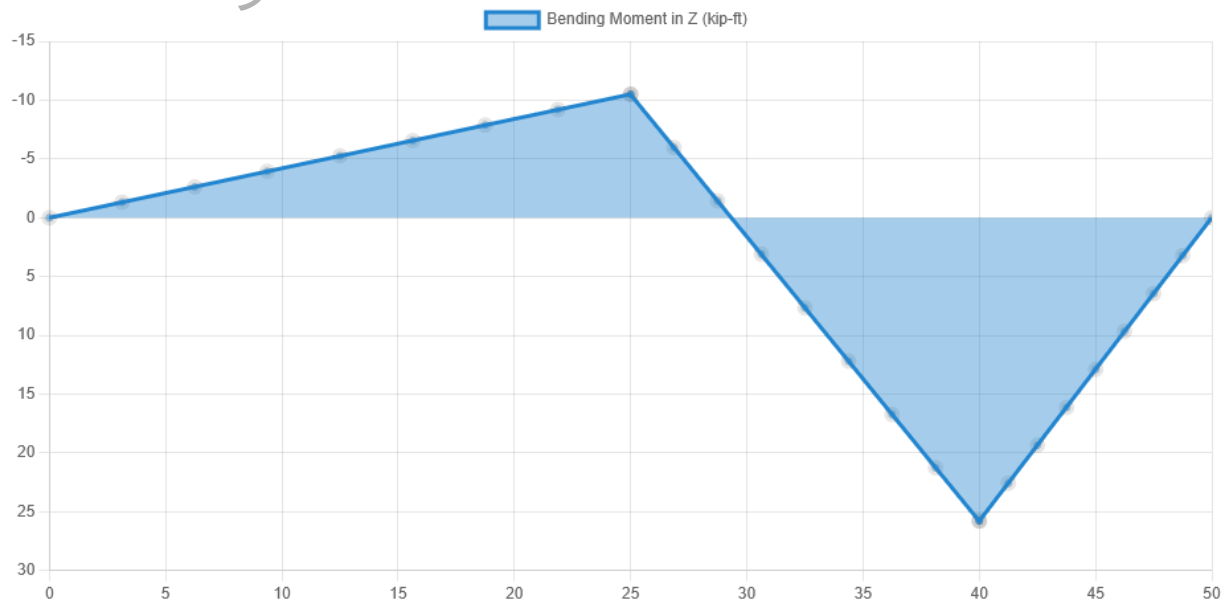
Result	Max	Min
Bending Moment	25.8 kip-ft	-10.5 kip-ft
Shear	2420 lb	-2580 lb
Displacement Y	0.039 in	-0.106 in

Stress Extremes

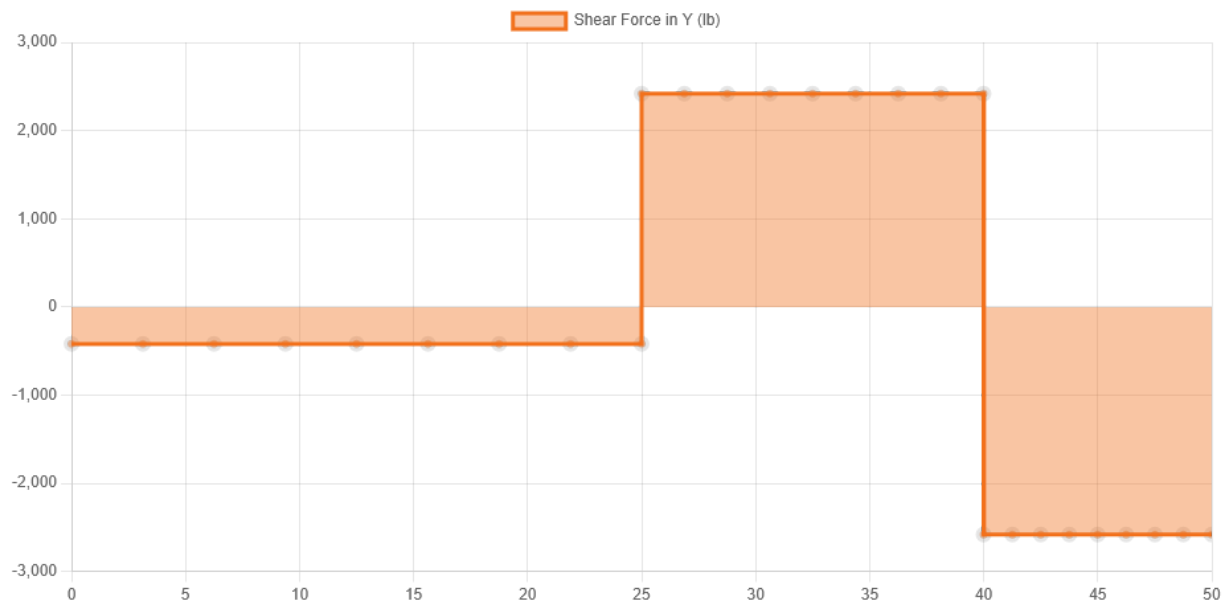
Result	Max (ksi)	Min (ksi)
Bending Stress	3.362	-3.362
Shear Stress Total	0.551	0.09
Max Combined Normal Stress	3.362	0
Min Combined Normal Stress	0	-3.362

DIAGRAMS

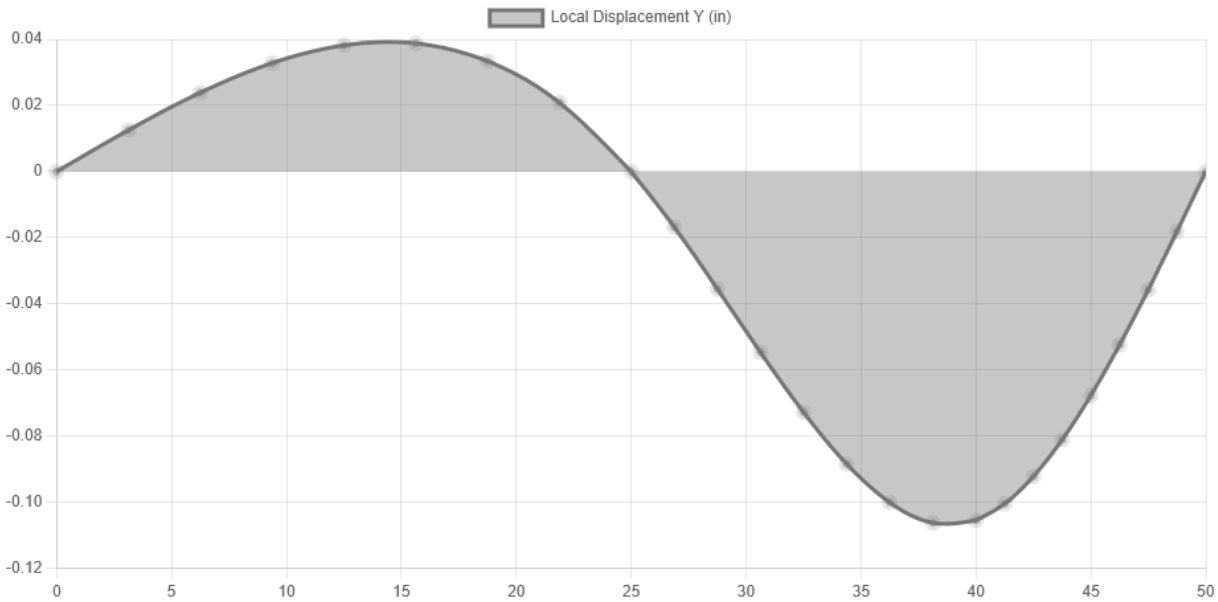
Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	0	15.625	25	38.125	50
Total Deflection (in)	0 ⓘ	0.039 ⓘ	0 ⓘ	0.106 ⓘ	0 ⓘ
Deflection/Span	-	L/7735	-	L/2824	-
ⓘ The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the model settings.					